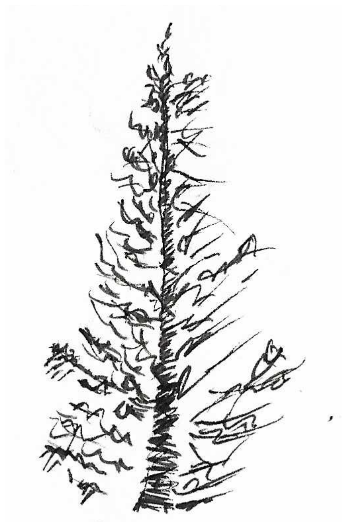
PDX Community Tails OS Workshop

v0.5



Contents

[2] "The Basics" Thread Modeling Fundamentals Håkan Geijer https://opsec.riotmedicine.net/downloads#threat-modeling-fundamentals

[3] https://tails.net/doc/first_steps/start/pc/index.en.html#boot-menu-key

[4] https://www.tomshardware.com/reviews/bios-keys-to-access-your-firmware,5732.html

- Signal (or other messenger app like Discord, ect.)
- Magic Wormhole
- Bitwarden Send.
- Warpinator (Winpinator for Windows)

13.2 Workshop Exercise

- Find a partner.
- Save the secret string you saved in KeePassXC to a file.
- Exchange secret files with your partner.

References

[0] "Motor Vehicle Traffic Crashes as a Leading Cause of Death in the United States, 2016, 2017", NHTSA. https://crashstats.nhtsa.dot.gov/Api/Public/ViewPublication/812927

[1] "Security vs privacy vs anonymity" pg 22. Turn Off Your Phone.org https://www.fightforthefuture.org/wp-content/uploads/2025/09/FullGuide-TurnOffYourPhone.pdf

Part I
Intro

1 Intention

Technology rewards the curious.

I am running this workshop to equip my community with tools to better protect its safety.

That said, it can be very easy to become overwhelmed and fatalistic, obsess over worst case scenarios, or become fixated on mythical "most secure" or "correct" tools and processes. (The only way to be perfectly secure is to talk to no one, do nothing of consequence, and never take a single risk).

We learn best when we are curious, playful, and consistently engaged. While reading a list of expert recommendations once probably makes you a little safer than nothing, you are far better served improving your safety skills a little bit every week: trying new tools, researching, and discussing it with people, especially people you don't necessarily agree with. If you find learning about computers or online privacy miserable, you aren't going to do any of that.

I will personally consider the workshop a success if everyone who attends learns:

- Something they think is cool.
- Something that makes them curious to know more.
- Something that equips them to be one modicum safer than they were before.

2 What Is Tails?

Tails (The Amnesic Incognito Live System) is a tool for mitigating digital surveillance and censorship.

It is a bootable Live USB. You plug it into a computer, and boot into a temporary separate operating system.

By default Tails will:

- Retain no data between sessions to avoid leaving traces of what you use it for.
- Route all internet traffic over the Tor network to provide privacy and anonymity while using the internet.

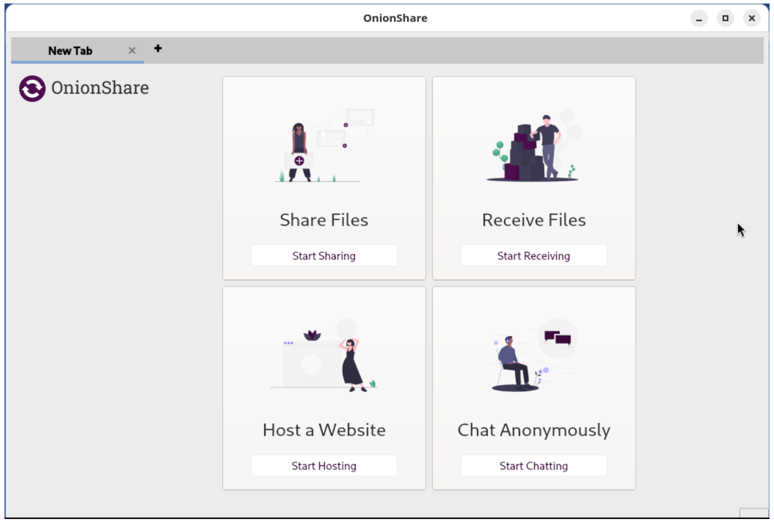
13.1 Sharing OnionShare URLs And Private Keys

Ways to share the OnionShare URL and or private key from within Tails OS.

Note: none of these options are end-to-end encrypted, but can be fairly anonymous.

- A throwaway or anonymous email address.
  - Tuta mail lets you create email addresses without an additional verification info. You just have to wait for a few days to complete verification.
  - Gurrillamail.com lets you create disposable email addresses.
- Riseup pad
  - You still have to send the URL/private key for the Riseup pad, but it can be pre-shared either in person or via a different means.
- Browser based chat

The following methods require either installing additional software into your Tails USB, or copying files to an extra USB to transferring to another computer that has this software, both of which present different kinds of security compromises.



- OnionShare can be opened under "Apps > Internet > OnionShare
- To share files in "Sharing" mode, click "Start Sharing"
  - Select a file.
  - Selecting "Always open this tab when OnionShare is started will keep your Onion address as long as your Tails session is running.
  - Selecting "This is a public OnionShare service" disables the private key.

3 Terminology

Some very quick definitions. A shared understanding of these terms makes us more effective when talking about digital privacy.

- Tor
  - Tor stands for "The Onion Router" and is a privacy preserving network. The easiest way to to access webpages over Tor is with the Tor browser. Tor routes all traffic through at least three encrypted relay nodes before being routed to its final destination. Traffic is encrypted in layers, with each relay node removing a single layer of encryption. This is designed to prevent any one node from knowing the source and destination of your web traffic.
- Onion Services
  - Onion services, (also called hidden services), are web services that can only be reached via the Tor network. Some services are only accessible via Onion services (the so-called "Dark web"), while other services have clearnet versions, while also offering an Onion service as a privacy preserving option for users. For example, here is the onion address for the search engine Duck Duck Go:

https://duckduckgogg42xjoc72x3sjasowoarfbgcmvfimaftt6t

wagswzczad.onion

  - Github repository of onion sites:
    https://github.com/alecmuffett/real-world-onion-sites

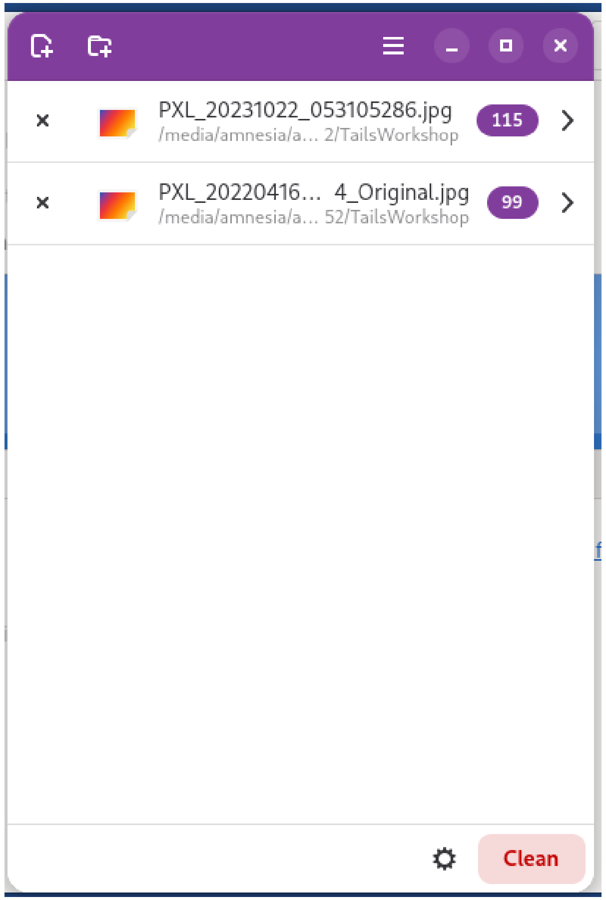
- Anonymity - Who is talking to who? Metadata.
- Privacy - What are they talking about?
- Security - How resistant is their communication medium to interventions that downgrade it in potentially several ways[1]:
  - Anonymity, unmask the participants.
  - Privacy, gain access to the contents of their communication.
  - Availability, take the medium down and make it unavailable for use.
  - Otherwise co-opt or use the medium for unintended purposes.

12.1 Workshop Exercise

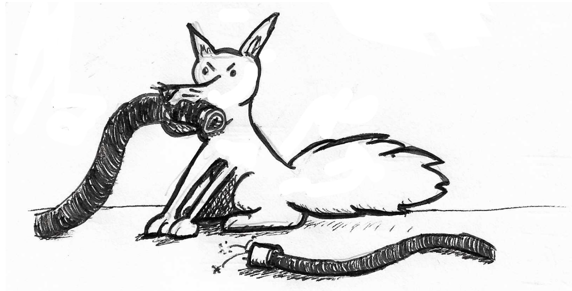
- Open the files you downloaded earlier in MetaData Cleaner
- View the metadata stored in the files.
- Clean the metadata
- View the file metadata after cleaning.

13 Using OnionShare

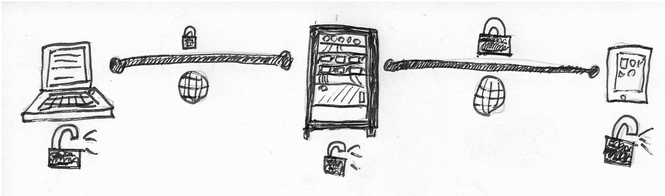
OnionShare is used for sharing files peer-to-peer. When you share a file, an onion address will be generated for the other party to open in TOR to access your share. Unless configured to disable it, you will also need to share a private key. It is advisable to share the URL and the key via two different channels.



- When you are ready to clean the data, hit the "Clean" button in the bottom right hand corner.

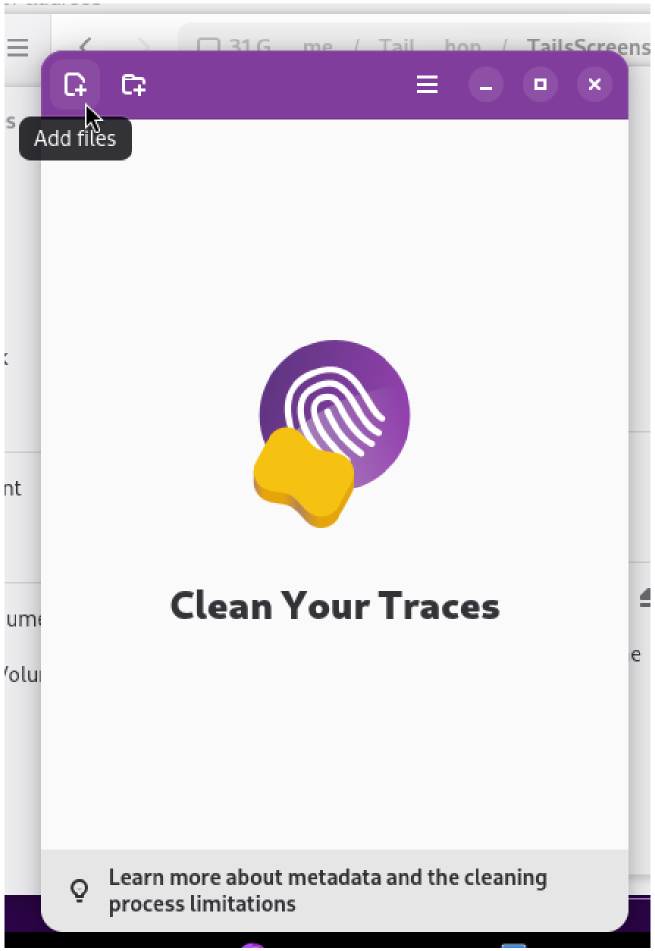


- Encrypted in Transit
  - Most websites use a technology called SSL, using the "HTTPS" prefix. This means the traffic is encrypted between the client and the server, and can't be read by the ISP or anyone else listening on the network. It can still be read by the server. While the traffic is private over the network to the server, it is not anonymous over the network, as the destination IP and DNS requests associated with the request are visible.
  - For example, Discord is a messaging app that is encrypted in transit.

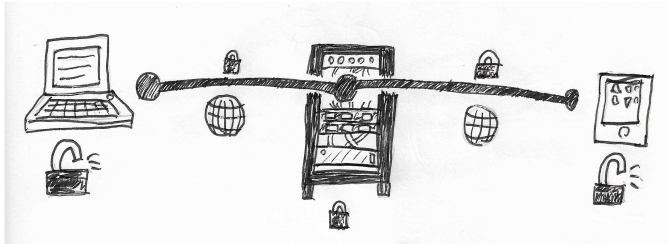


- End-To-End Encrypted (E2EE)
  - Services that are end-to-end encrypted mean the data is encrypted in transit, and also encrypted on the server. This is usually used to refer to messaging apps where only the intended recipient can decrypt the message. Typically a server is involved, routing the encrypted data from one client to another. End-to-end encrypted services offer privacy for the contents of messages, but don't usually offer anonymity, as metadata like who is talking to who, and when, is still visible to the server.
  - Signal is end-to-end encrypted, but not anonymous, and requires users to provide a phone number to use the service.

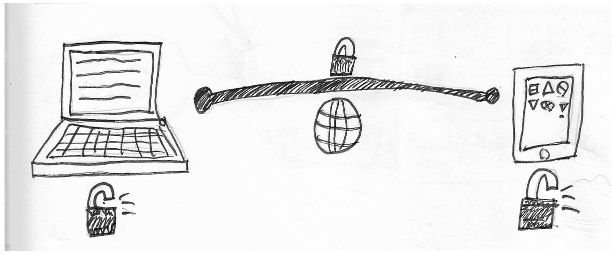
- Open a file or folder by pressing the add buttons in the top left of Metadata cleaner's window.
- You can view the medatadata associated with the file by clicking on the arrow to the right of the file name.



- Open Metadata cleaner under “Apps > Metadata Cleaner”.



- Peer-To-Peer (P2P)
  - Services that are peer-to-peer don’t involve a central server at all. Instead individual clients connect directly to one another. Usually this means both parties must be online at the same time, and have a way of pairing their devices either in person at a previous point, or via a different, “out-of-band”, communication method.
  - Briar is a peer-to-peer messenger app.
  - SimpleX and Delta Chat are mostly peer-to-peer, relying on servers to connect peers but take steps to run multiple servers by different parties and minimize the amount of metadata any individual server has.



- Difficulty Differential
  - Mitigations that are easy for me but hard for an attacker are the most valuable. Some mitigations will be more like 50/50, these are less useful but not useless depending on your needs.
- Risk
  - Almost everything we do involves some level of risk. In some ways, we are experts at taking risks. In other ways, humans are pretty terrible at it. It can be challenging to accurately gauge risk or have productive conversations about it. Below is one definition of risk I find helpful for talking about and thinking about taking risks.
  - Risk = Cost of bad thing * Likelihood of bad thing
  - Example:

12 Using MetaData Cleaner

MetaData cleaner is an application for scrubbing metadata from files. If you intend to share documents, images, and other files, removing metadata is one step that can preserve anonymity. Note that MetaData cleaner can remove a lot of metadata, but it isn't perfect, and there is some metadata it may miss. Also note that sharing images or video anonymously is extremely difficult. Locations can be determined from landmarks or other features by even skilled hobbyists, and images of specific clothing or other features could be used to identify you if correlated against other images.

using EFF's diceware process. The passphrase should be at least six words long. If using KeePassXC, click on the little dice icon to the right of the "Enter password" field. Change to the "Passphrase" tab, toggle the visibility (the eye icon at the top right), and make sure the word count is *at least* six words. Although it might be tempting to change the word case or word separator, adding more words is a more effective way to make the passphrase stronger. Make sure to memorize the passphrase or write it down somewhere safe since that is the only way to unlock the database file.

- After setting the passphrase, you will be asked where to save the database. Save it to your LUKS-encrypted USB with the default file name ("Passwords.kdbx"). It is recommended to keep the database somewhere obvious and not to move it. Once the database is saved, you can add secrets (eg. passwords) to it!

- Note: you can access this database from any device that has KeePassXC (or KeePassDX or KeePassium) if it has access to the file (or a copy of it) and the passphrase.

11.1 Workshop Exercise

Create and store a secret string for sharing with a partner in a future exercise.

    * Risk of getting in a car crash: Cost: Pretty high, ranges from injury or death depending on the kind of accident.
    * x
    * Likelihood: Depends on your age, lifestyle, and other factors. In the United States it ranked as the 13th leading cause of death among all causes in 2016 and 2017.[0]
    * = Risk depends on individual factors.
  - Some risks have a harder to quantify likelihood than others. The likelihood can also change over time or from person to person. Similarly, the impact of a risk can differ from person to person and sometimes be difficult to estimate. All of these factors can make estimating risk a challenge.

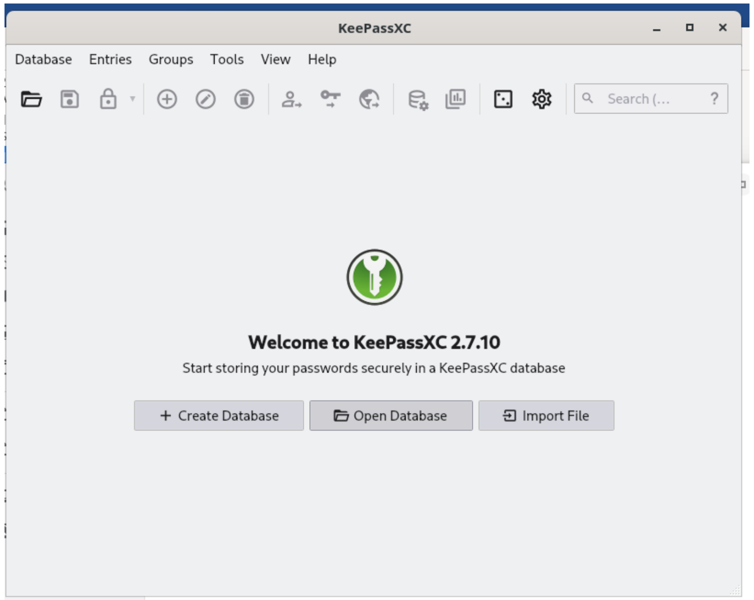
- Risk Tolerance

  - Everybody's tolerance for risk will be different, both in general and with regards to specific risks. People's individual health, privilege, family or other responsibilities, and other life factors can all influence how tolerant to accepting risk they are. While getting doxxed isn't fun for anyone, some people may find a chance of getting doxxed an acceptable risk, while for others it may severely impact their ability to find work and otherwise completely derail their life. When engaging in or discussing risk-taking behavior, it is important to be honest with yourself about the level of risk you can tolerate and endeavor to not over or under estimate. It is also important to be mindful of other's risk tolerance when discussing risk-taking behaviors. While its healthy to have a dialogue, even a critical one, about realistic risk tolerance, be wary of pressuring others into accepting more or less risk, or assuming that everyone has the same risk tolerance you do.

4 Why Should I Use Tails? Things Tails Can and Cannot Do.

4.1 What Tails Does Well:

- Keeps web traffic anonymous and private from internet service providers or other people on the network.
- Keeps you anonymous from the website you are visiting, provided you don't supply identifying information.
- Provides decent security against most threats.



- If needed, attach the intended storage device then open KeePassXC via "Apps > Accessories > KeePassXC". Click the "Create Database" button (or "Database > New Database") to create a new database file. Continue through the "General Database Information" and "Encryption Settings" screens without changing the defaults. The database name should be "Passwords" with no description. The database format should be "KDBX 4 (recommended)".
- Finally, generate a passphrase - either within KeePassXC or

11 Setting up a Password Manager with KeePassXC

KeePassXC is a password manager that can be used to storing passwords. Generating random passwords stored in a secure location is safer than re-using passwords. Unlike cloud-based password managers, KeePassXC stores passwords in a file ("Passwords.kdbx") that needs to be saved somewhere safe. Although it is possible to create "Persistent Storage" inside Tails, this is not recommended. We recommend using a separate LUKS-encrypted USB to store this file.

4.2 What Tails Doesn't Do Well:

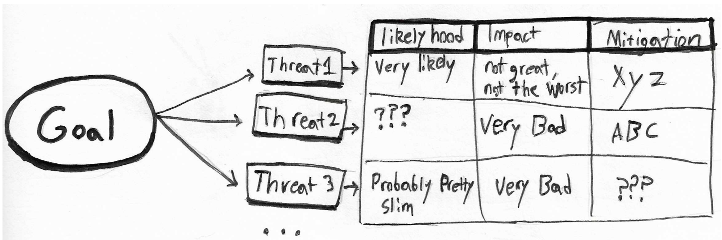
- Hide from your Internet Service provider you are using Tor.
- Hide from the website you are visiting that you are using Tor.
- Protect your anonymity if you do something that unmasks you.
- Provide security against threat actors with nation-state level resources.

4.3 Threat Model

Whether Tails is an appropriate tool for you depends on what you are trying to do and your threat model.

- A Threat Model is a structured way of thinking about your goals, obstacles, and people who may not want to you accomplish those goals. Threat modeling is a continuous practice, you should revisit your threat model as time goes on and update it as new information becomes available about your threat actors, additional mitigations, and the effectiveness of your existing mitigations. Several formalized frameworks exist. I'll be using the framework outlined in the Zine "Threat Modeling Fundamentals" by Håkan Geijer. [2]
  - What are our goals?

    * Who or what might oppose or complicate those goals? What kinds of tools do they have? What risk do they pose? What the likelihood they would interfere? What would the actual impact be if they did?
    * What can we do about it? Do we have a mitigation that brings risk down to negligible levels? What about mitigations that reduces risk a little, but still has a tangible likelihood? Or is this a risk we are forced to accept?
    * Is our threat model well informed? Is it effective?

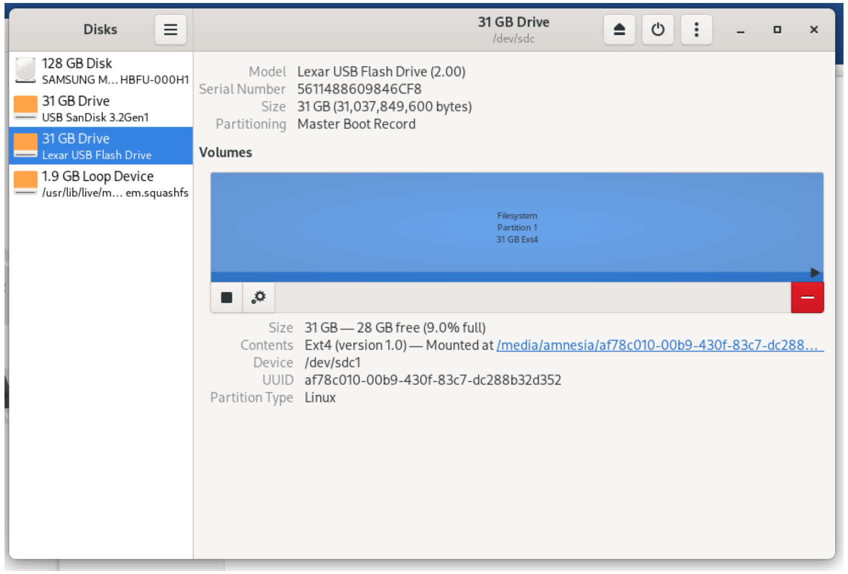


- Below are additional resources on threat modeling
  - Frameworks:
    * HARMS framework focuses less on technical treats and more on human impact:
      · https://arxiv.org/html/2502.07116v1
    * This zine explores threat modeling in more detail and is tailored towards activists:

- Finally, we create the encrypted partition by selecting the device-to-be-encrypted then clicking the + button to create a new partition. The partition creation assistant will guide you through some options. Below are our recommendations:
  - Partition Size: Cover the entire size of the device.
  - Volume Name: Optional, but could be useful to identify it.
  - Erase: Optional, but not harmful at this stage.
  - Type: Select "Internal disk for use with Linux systems only (Ext4)" and "Password protect volume (LUKS)".
  - Password: Enter a passphrase of at least six *random* words. Don't just make up words! If needed, look up EFF's diceware word list for help.
  - After entering a sufficiently random passphrase, click "Create" to start the LUKS encryption.

10.1 Workshop Exercise

Set up a LUKS encrypted drive.



- Before continuing, make sure you are OK with permanently deleting whatever data was already on the device. When you are ready, select the device-to-be-encrypted (click on it to highlight it) then click the button that looks like three dots in a vertical line in the title bar. Select the "Format Disk" option to erase all existing partitions on the device. In the "Format Disk" dialog, choose "Overwrite existing data with zeroes" in the Erase menu. When ready, click "Format" twice (second time confirming that you are sure). This may take some time.

      · https://opsec.riotmedicine.net/downloads#threat-modeling-fundamentals

  – Research

    * Online database of threats and mitigations, tailored towards activism:

      · https://www.notrace.how/threat-library/

    * Talking with people, researching online (potentially inside of Tails), testing out and experimenting with mitigations yourself.

Part II
Creating and Setting up Tails

5 Downloading the Tails ISO

- Navigate to Tails.net and download the Tails ISO.
  – https://tails.net/install/download/index.en.html
- Verify the download using the verification tool on the download page.

6 Flashing the Tails USB

- For Windows use Rufus
  - https://tails.net/install/windows/index.en.html
  - https://rufus.ie/en/
- For Mac use Raspberry Pi Imager
  - https://tails.net/install/mac/index.en.html
  - https://www.raspberrypi.com/software/
- For Linux you can use GNOME Disks or ‘dd‘
  - https://tails.net/install/linux/index.en.html

7 Booting From the USB

There are a few ways to boot from the USB, depending on the kind of laptop you have.

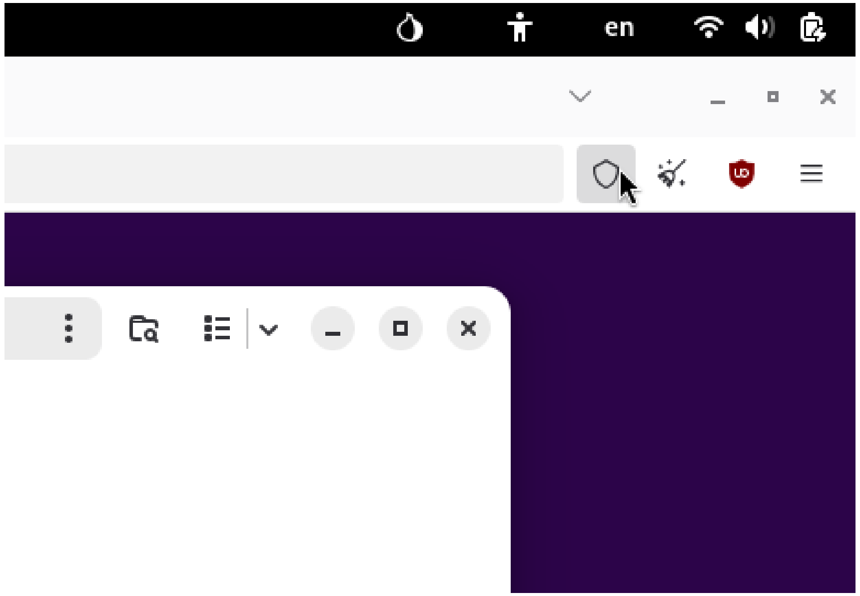
7.1 Booting from within Windows

If you are using Windows 8 or 10 you can start the process of booting to Tails from within Windows. https://tails.net/doc/first_steps/start/pc/ restart

10 Creating a LUKS Encrypted Drive

LUKs stands for “Linux Unified Key Setup”, and is a standard for encrypting hard drives. In Tails, you can create an encrypted external flash drive for storing files in. Storing files for a specific project inside a dedicated encrypted flash drive lets you isolate projects. If your Tails session becomes compromised, data stored on external drives and not attached to the system will be safe.

- Start with the device-to-be-encrypted unplugged. First, open the Disk utility by clicking "Apps > Utilities > Disks". The Disks utility will list all current storage devices on the left side of the screen. Next, plug in the device-to-be-encrypted and look for it in the Disks utility. Before proceeding, make sure the description matches what you’d expect (brand, size, etc).



9.5 Workshop Exercise

- Access the Onion Service provided by the organizer and download the files at the service.
  - The onion address for the files will be posted to a RiseUp Etherpad
  - https://pad.riseup.net

1. Make sure the Tails USB is plugged in.
2. Click Start
3. Hold "Shift" while selecting "Power->Restart"
4. The machine will restart, but will boot to a menu of diagnostic options. Select "Use a device".
5. If your Tails USB stick comes up in the list of options, select it.
6. Otherwise, if "Boot Menu" is an option, select that.
7. If neither of those options are present, you will have to try launching Tails using the Boot Menu Key.

7.2 Booting with the Boot Menu Key

1. Plug in the Tails USB.
2. Shut down the computer.
3. Computers from various manufacturers use different buttons to enable the user to bring up a boot menu when the computer starts up. Refer to the table below for common buttons

associated with different manufacturers. You may have to experiment with different buttons to find the right one for your computer.

4. Start the computer, repeatedly mashing the boot menu key for your computer. If successful, a menu of boot options will come up. You may fail the first few attempts if you miss the window or if the boot menu key is an unexpected key.

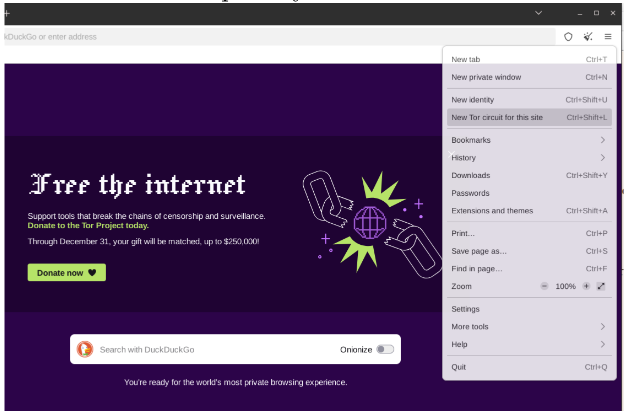
9.4 Tor Browser Security Settings

The Tor browser has settings that allow you to enable or disable various levels of JavaScript features. A skilled attacker could use malicious JavaScript on a web page you visit to compromise your Tails session security. Changing the "Security Level", disables Javascript, which is a strong mitigation against these kinds of attacks, but can make some websites nonfunctional.

To change the "Security Level", click the shield in the upper right hand corner, to the right of the search bar. Click "Settings" in the pop up that appears, and then select between "Standard", "Safer", and "Safest" security levels.

blocked by the site you are trying to visit. To request a new Tor circuit, select the options menu in the top right hand corner of the Tor browser, and select "New Tor Circuit for this Site".

Note, if you are trying to start a new session to preserve anonymity, it is safer to restart Tor completely or restart Tails.



9.3 Preserving Anonymity

Tails can't preserve your anonymity online if you supply information that identifies you. If you are trying to remain anonymous during your session, don't log into any accounts that you use on other devices. If you create an account inside of a Tor session, don't log into that account outside of Tor.

Manufacturer	Key
Acer	F12, F9, F2, Esc
Apple	Option
Asus	Esc
Clevo	F7
Dell	F12
Fujitsu	F12, Esc
HP	F9
Huawei	F12
Intel	F10
Lenovo	F12, Novo
MSI	F11
Samsung	Esc, F12, F2
Sony	F11, Esc, F10
Toshiba	F12
Others…	F12, Esc

[3]

7.3 Changing the Boot Order in the BIOS/UEFI menu

1. If the other approaches fail, you can access your computer BIOS or UEFI and change the order of boot devices.

2. To bring up your BIOS/UEFI, a specific button needs to be pressed when the machine starts. Below is a table of the common buttons for bringing up the BIOS/UEFI. [4]

   (a) ASRock: F2 or DEL

   (b) ASUS: F2 for all PCs, F2 or DEL for Motherboards

   (c) Acer: F2 or DEL

   (d) Dell: F2 or F12

   (e) ECS: DEL

   (f) Gigabyte / Aorus: F2 or DEL

   (g) HP: F10

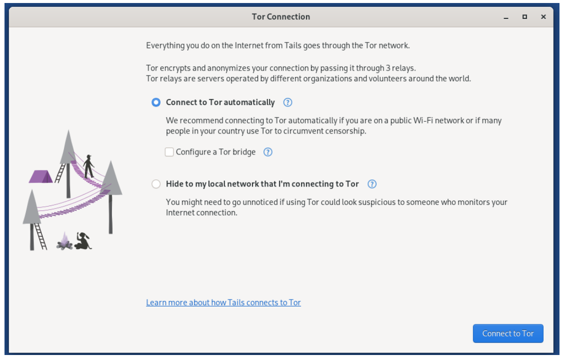
   (h) Lenovo (Consumer Laptops): F2 or Fn + F2

   (i) Lenovo (Desktops): F1

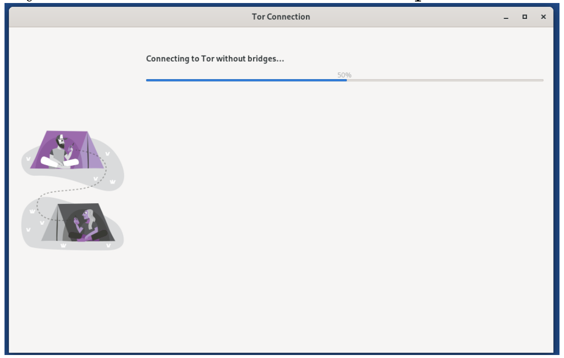
   (j) Lenovo (ThinkPads): Enter then F1.

   (k) MSI: DEL for motherboards and PCs

   (l) Microsoft Surface Tablets: Press and hold volume up button.



Hit “Connect” and the system will begin setting up your Tor circuit. It may take several minutes to complete.



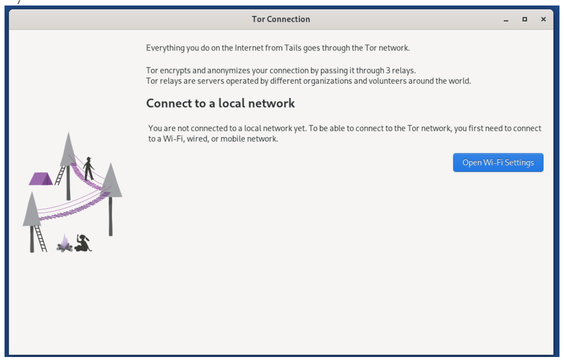
9.2 Sites that Block Tor

Some sites block access from the Tor network. One thing you can try is to use a new Tor circuit. Not all Tor exit nodes may be

9 Using Tor

9.1 Connecting to Tor

When you first launch Tor, you will be presented with a screen for configuring your Tor session. If you are not connected to the Wifi or a network, do that first.



In the following menu, select "Connect to Tor Automatically". You will be given the option to connect with a Tor bridge. Tor bridges are essentially a VPN for the first hop of the Tor circuit, providing a way to access the Tor network in countries that block it, or if you want to hide from your ISP you are using Tor. This may or may not be necessary given your threat model and risk tolerance.

(m) Origin PC: F2

(n) Samsung: F2

(o) Toshiba: F2

(p) Zotac: DEL

3. Power on your computer and hit the button over and over, this should bring up the BIOS/UEFI for your machine. You may have to experiment with different buttons to find the right one.

4. Once in your BIOS/UEFI, there should be an option to manipulate the order of default boot devices. You will want to make USB the highest priority boot device.

8 Selecting Boot-up Options

When you launch Tails, you will be presented with a menu of options for configuration including Language and keyboard layout.

8.1 Persistent Storage

The option for "Persistent Storage" allows you to setup up a dedicated encrypted partition where files can be saved across Tails sessions. Saving files to a separate encrypted flash drive offers more flexibility and security, however Tails can save settings like WiFi passwords or custom installed software to the persistent storage. Setting up persistent storage is a trade off of security in exchange for convenience. If your Tails session is accessed remotely by an attacker, stored settings could be used to help identify you.

8.2 Administration Password

Under "Additional Settings", there is an option for "Administration Password", when enabled, this allows you to set an admin or "root" password for your session, and run a terminal with admin privileges. Admin privileges can be used to install additional software do other customizations to your Tails session you would otherwise be locked out of. When you restart Tails, the Admin password is disabled and reset.

Using the Administration password is another security trade off, one that comes with bigger risks than persistent storage.

When an administration password is unset, it is much more difficult for an attacker to potentially install malware or otherwise tamper with your Tails session.

Furthermore, one of the ways Tails provides anonymity is by offering a standardized sandbox, one that is indistinguishable from anyone else running Tails to any attacker able to collect identifying information about your machine. Installing custom software breaks that standardization and makes your Tails USB recognizable as a unique instance of Tails, which could be used to help identify you.

Finally, extra software presents additional potential vectors for attack. The software bundled with tails is audited. Bugs that can be exploited by an attacker are still found sometimes, but the level of scrutiny is fairly high, driving down the risk of exploitable bugs being present, and making any bugs that are found more valuable and therefore more less likely to be used on a regular basis. Any extra software you install may or may not have the same level of auditing and scrutiny. Even if it is well audited, all software has bugs, and adding software with very few bugs is still more potentially exploitable bugs than existed before.

That being said, if your use case calls for installing and running a specific piece of software, you can do it by enabling the Administration Password and then opening the Synaptic Package Manger or installing from the command line.

Part III
Using Tails

Tails has a handful of tools for doing basic computing tasks, the Tor browser for web surfing, a basic Office suite, an offline password manager, sound and video players, and some system tools.